Project Design Document

1. Design Intent

- The experience we are trying to create is a Harry Potter themed interactive table game based on the Triwizard Tournament. The player goes through three trials which includes different types of games which include themes from the original tournament to keep the story telling consistent and if they complete all three, they're crowned Triwizard Champion.
- It's for all the Harry Potter fans, but realistically it's designed in a way that even if a person who has no prior idea or experience of the harry potter theme can equally be immersed into the experience and have fun. It does not solve a real world problem, for us it was about creating something that feels like a proper game experience, not just a series of puzzles. The main ideating challenge was making each trial feel different, use different sensors and outputs, and be enjoyable to play rather than just technically functional and focusing on the visuals of what would physically be happening if a player were to lose or win rather than only focusing on the game technicalities.

2. Inspiration & References

- The main inspiration is obviously from the Triwizard Tournament from Harry Potter and the Goblet of Fire ,specifically the three tasks which include the golden egg from the dragon, the maze and the gillyweed. The game maps each of these into something a player can physically interact with on a table.

- The main idea of the goblet is a direct inspiration from the film, we would like to replicate the effect by using tinted blue film and a fan at the bottom which would recreate the roaring flame .

- The mechanism of the dragon storing its egg inside its mouth comes from a few inspirations we found for box storages. The neopixel oxygen cylinder in Trial 3 was inspired by those countdown timer bars you see in underwater video games which keeps the trial true to the lake theme. A visual that's depleting in real time creates genuine pressure without needing a screen or numbers to show timer

- What we learned from looking at similar interactive prop builds made for any adaptations is that the physical theming matters just as much as the electronics. If it looks the part, players engage more like a cardboard dragon with a moving mouth is more exciting than a perfectly coded LED strip on a plain board. It would be immersive and engaging if the visuals match the depth of the storytelling

2. User Journey

Step 1 — The Goblet of Fire- The player walks up to the table and sees the Goblet of Fire prop glowing blue, with the fan making the film paper flicker like flames. They write their name on a piece of paper, fold it, and drop it into the goblet. It falls into the bowl at the bottom, blocking the LDR. The system detects this, and this starts the game by showing a laser cut screen showing the words “you are chosen”. The game has officially begun.

Step 2 — Trial 1, The Dragon A neopixel strip runs along the body/spine of a cardboard dragon. The pixel moves back and forth like a heartbeat pulse , this represents the dragon's heart. There's a target zone marked on the dragon's body. The player has to watch the pulse and press the button at exactly the right moment to land in the zone, like stopping the dragon's heart. If they get it right, the servo motor activates, pulling a string attached to the dragon's upper jaw ,the mouth swings open on its cardboard hinge and reveals the golden egg inside. If they miss, the dragons eyes turn red by red leds placed behind its eyes, and 2 seconds later ,the pulse continues and they try again. They have three tries before the game resets and the player is eliminated.

Step 3 — Transition The player is then handed an another card which has the instructions for the next trial . When the player presses the ready button, Trial 2 begins.

Step 4 — Trial 2, The Rune Sequence Six symbols are laid out with individual LEDs. The system lights them up one by one in a random sequence, like say, 2, 4, 6, 5, 1, 3. The player watches the full sequence, then has to replicate it using the corresponding push buttons. Once the user is done , if they match the full sequence, the game progresses or they get three tries. If they use all three lives, the game ends and the player is eliminated.

Step 5 — Trial 3, The Black Lake In front of the player is a bed of tissue paper gillyweed with three LEDs glowing inside it — these represent the three weeds still left to harvest. To the side is a tall cylinder wrapped in a neopixel strip, fully lit at the start — this is the oxygen supply. It's slowly depleting. system picks a random target height ,say 15 cm and a corresponding LED label lights up to show this to the player. The player hovers their hand over the ultrasonic sensor at that height to "harvest" the weed. If they get it within the tolerance window, one of the bed LEDs turns off (showing weed harvested) and they move to the next target. If they get it wrong, the oxygen depletes faster as a penalty. The goal is to harvest all three before the neopixel cylinder goes completely dark and oxygen runs out.

Step 6 — Champion All three trials complete. A trophy prop rises from behind the game table driven by the stepper motor and celebration LEDs fire across the whole board. The player is awarded as a Triwizard Champion.

3. Success Criteria

Functional (e.g., LED responds in <1 sec)

- LDR in the goblet detects paper drop within 1 second and triggers the start sequence reliably every time
- Neopixel heartbeat in Trial 1 runs at a consistent speed and the target zone registers a button press correctly with no delays in input taken
- Servo motor opens the dragon's mouth fully without the string snagging or loss in tension
- Trial 2 sequence plays fully before accepting input, and button presses register accurately with no double counting
- Ultrasonic sensor in Trial 3 averages readings and accepts a correct distance within ±2 cm tolerance
- Neopixel oxygen cylinder depletes in real time and responds to wrong answers with a faster drain in oxygen
- Trophy stepper motor rises smoothly on game completion with no disruptions
- Full reset button resets all game state cleanly for the next player

- *User experience (easy to use, intuitive)*

- Each trial feels physically and thematically different from the others
- Transitions between trials feel like events, not just a list of games
- The difficulty is fair , not too easy to feel pointless, but not so hard that players get stuck and disengage
- *Reliability*

- The system should survive a full demo day of repeated playthroughs without needing a restart or repair

4. System Overview

ESP32 microcontroller controls all the sensors, LEDs, motors, and game logic.

The game is broken into different states- idle, trial 1, transition, trial 2, trial 3, victory and the ESP32 moves between them based on sensor inputs and player actions.

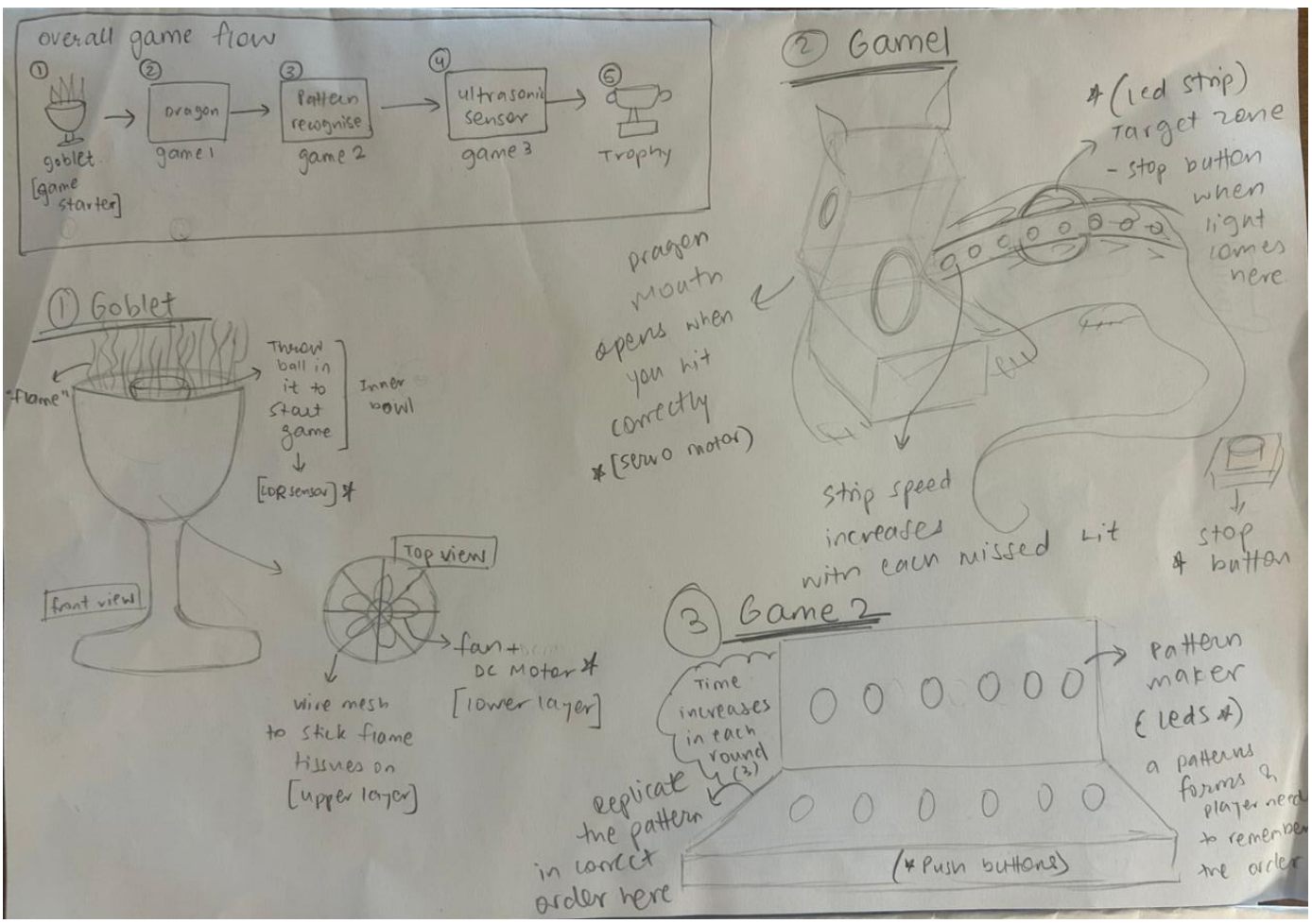
The main physical sections of the build are the Goblet of Fire prop, the dragon with the egg reveal, the symbol panel for the sequence trial, the gillyweed bed with ultrasonic sensor, the neopixel oxygen cylinder, and the trophy reveal mechanism at the end.

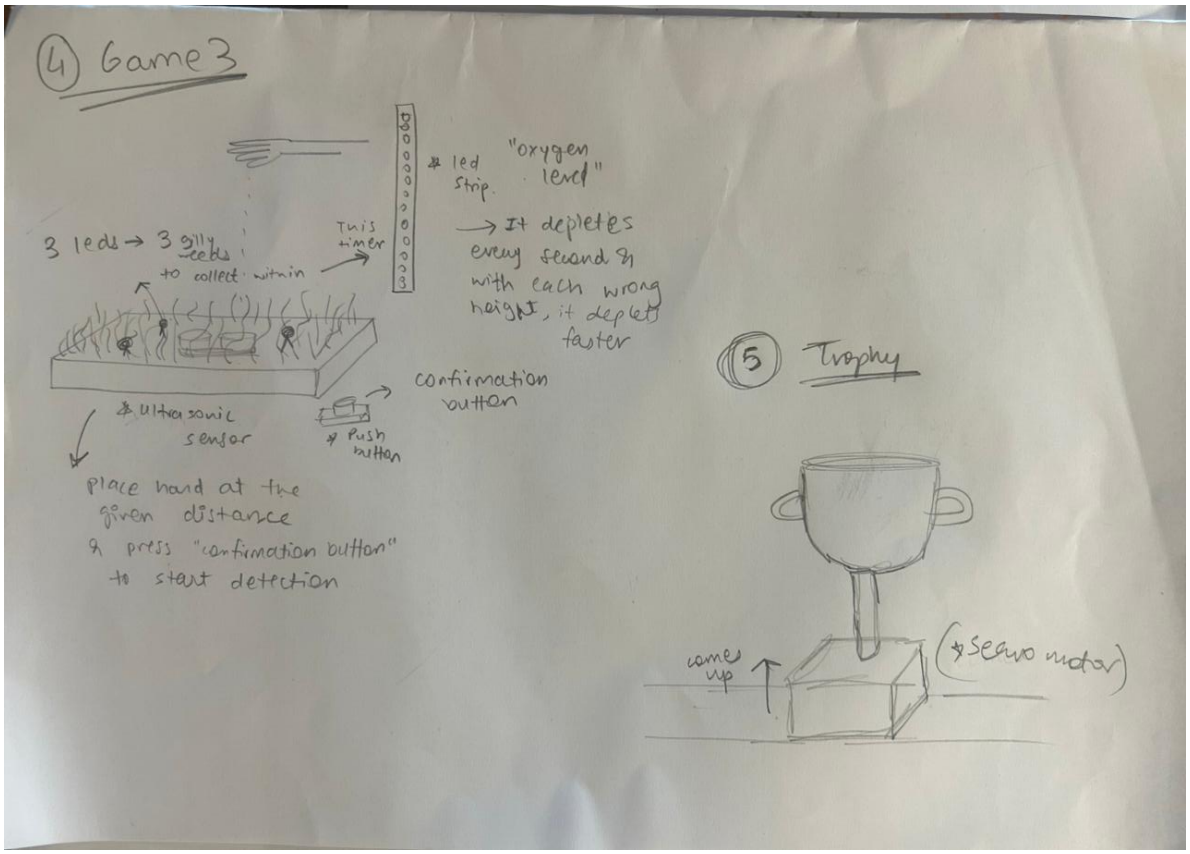
7. Input / Output Mapping

Input	Process	Output
Paper dropped in goblet	**Detect light drop below threshold → triggers game start**	**Display led turns on**
Button press during heartbeat (Trial 1)	**Check if neopixel position is within target zone**	**Correct: servo pulls string, mouth opens / Wrong: pulse continues**
Ready button pressed		**Trial 2 sequence begins**

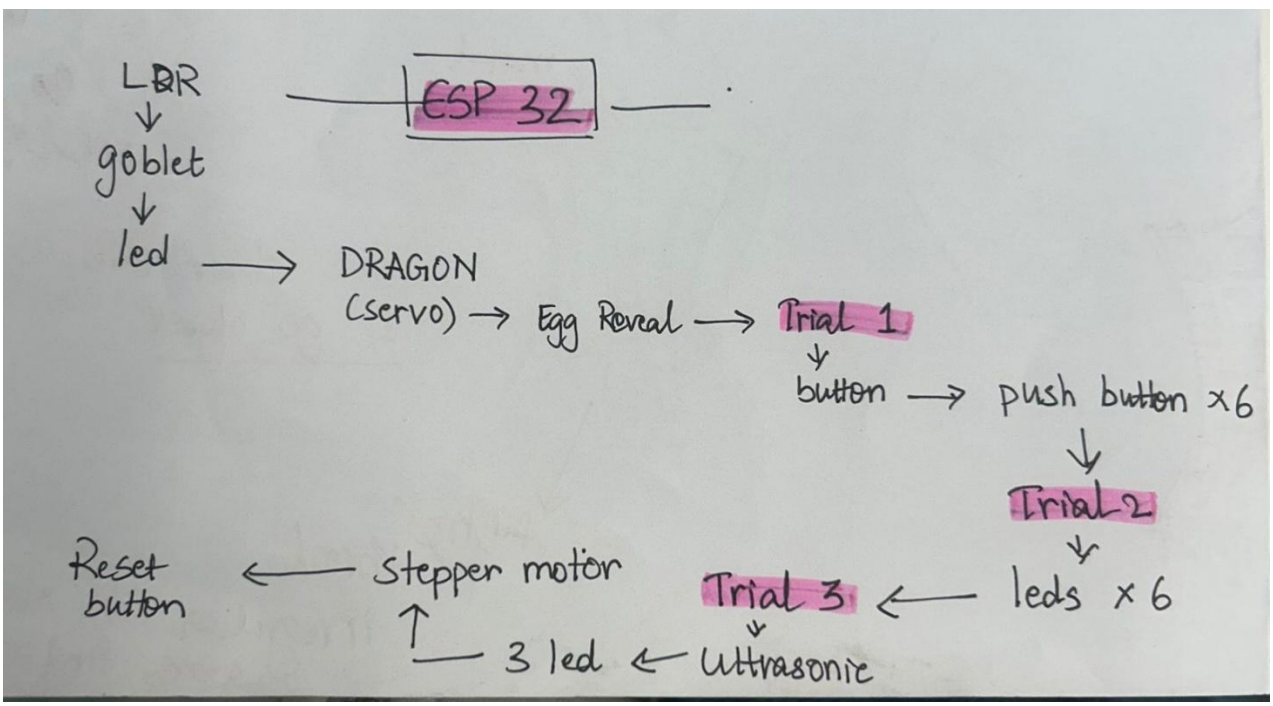
Push buttons pressed (Trial 2)	**Compare input sequence to stored sequence**	**Correct sequence: move to trial 3. Wrong : lose a life, red light**
Hand height over ultrasonic sensor (Trial 3)(3 times)	**Compare to target height ± 2 cm**	**Correct: bed LED off one by on** **Wrong: oxygen depletes faster**
Victory state reached	**Activate stepper**	**Trophy rises**

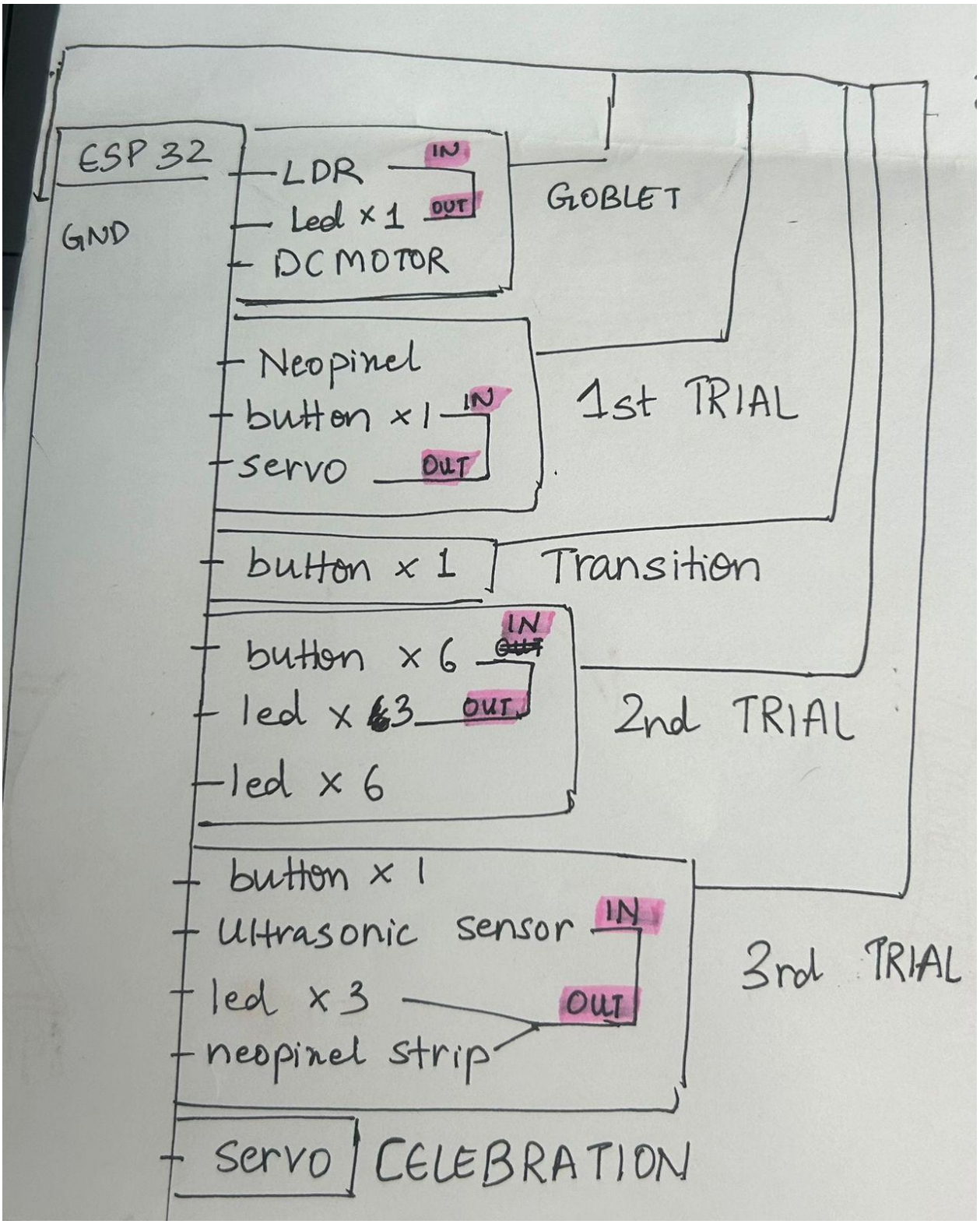
8. Sketches & Concepts





9. Final System Architecture (Labelled)




ESP 32
GND
LDR
IN
Led x 1
OUT
DC MOTOR
GOBLET
Neopixel
button x 1
IN
servo
OUT
1st TRIAL
button x 1
Transition
IN
button x 6
OUT
led x 3
OUT
led x 6
2nd TRIAL
button x 1
Ultrasonic sensor
IN
led x 3
OUT
neopixel strip
3rd TRIAL
servo
CELEBRATION